

Dashboard Challenge

A critical part of SSMIF Quant that glues all the other teams together is our web interface and end-to-end financial data forecast pipeline.

Your job is to create a web service that visualizes backtest results of trading algorithms on any symbol. Your application should comprise of (a) a database, (b) an ingester script that fetches data to store in the database, (c) a backend server that runs a backtest for any trading strategy on that data, and (d) a frontend that submits backtest requests to the backend, and displays its results.

What is expected from your submission is that the frontend will prompt us to set the start date/end date of the backtest, in addition to any parameters to the trading strategy you may want users to specify. The backend will then run the trading strategy on the provided timespan, producing a list of trades the strategy made, the profit/loss, annualized return, max drawdown, and win probability of the strategy to be displayed on the frontend. Additionally, the ingester script should be runnable by itself, but if some data is not present in the database for backtest requests submitted by the frontend, the backend will call the ingester script to fill the database with required data before starting the backtest.¹

For this challenge, it doesn't matter what symbol or strategy you decide to use: any stock, forex, options, crypto symbol is acceptable, and any strategy, as complex as a neural network or as simple as a constant price threshold² is acceptable, as long as it is runnable by your graders. Only one stock symbol and only one trading strategy is required, but multiple are recommended. We will be manually changing your hardcoded strategy(ies) to make sure your frontend will reflect the changing backtest results. You will not be graded on the quality of your trading strategy.

If you are selected to continue the application process, you will be expected to explain the challenges you have encountered while developing this application. Your code should be well documented and maintainable, following the common conventions of your programming language framework, and database of choice. Lastly, your work will be submitted in a git repo compressed in a zip archive.

1.1 Requirements

- Any database. (PostgreSQL, ClickHouse, etc.)
- A standalone ingester script for required symbols.
- A fullstack app that displays every trade that the strategy made over its backtest, and useful trading metrics: at least the profit/loss, annualized return, max drawdown, and win probability.
- A README.md doc that explains your trading strategy, allows graders to reproduce your backtest, and how to deploy your code.
- A Dockerfile.

¹E.g. if we make a backtest request for 9/1 to 9/30, but data for 9/29 is not present in the database, the backend will call the ingester script to fill the database with 9/29 missing data before running the backtest.

²One of the simplest trading strategies is the constant price threshold, to buy when the price breaks from below to above some predetermined constant value, and to close that trade after some time. This is easily debuggable, easily reproducible, and easily programmable. This strategy has two parameters you should let the user provide, which are the price threshold(s) to take the trade, and the time before the trade is closed.